

Maple Finance A-4

Security Audit

February 7th, 2025

Version 1.0.0

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Introduction

This document includes the results of the security audit for Maple Finance's smart contract code as found in the section titled 'Source Code'. The security audit was performed by the Macro security team on February 6th, 2025.

The purpose of this audit is to review the source code of certain Maple Finance Solidity contracts, and provide feedback on the design, architecture, and quality of the source code with an emphasis on validating the correctness and security of the software in its entirety.

Disclaimer: While Macro's review is comprehensive and has surfaced some changes that should be made to the source code, this audit should not solely be relied upon for security, as no single audit is guaranteed to catch all possible bugs.

Overall Assessment

The following is an aggregation of issues found by the Macro Audit team:

Severity	Count	Acknowledged	Won't Do	Addressed
Informational	1	-	-	1

Maple Finance was quick to respond to these issues.

Specification

Our understanding of the specification was based on the following sources:

- Discussions on Telegram with the Maple Finance team.

Source Code

The following source code was reviewed during the audit:

[syrup-utils-private repository](#)

- **Commit Hash (Initial):** c367a2e4569131c37075e0f314ad21a5e7fddb9e
- **Commit Hash (Final):** c367a2e4569131c37075e0f314ad21a5e7fddb9e

[open-term-loan-private repository](#)

- **Commit Hash (Initial):** f61d459a09a97d836405afb10db451f26579ebe0
- **Commit Hash (Final):** f61d459a09a97d836405afb10db451f26579ebe0

[fixed-term-loan-private repository](#)

- **Commit Hash (Initial):** 61696225a133daeabeec374c23f4274d64a1996a
- **Commit Hash (Final):** fc08a0425a86c4fa916b13750bbc547eeb3315c5

Specifically, we audited the following contracts within these repositories:

Source Code	SHA256
syrup-utils-private/contracts/MapleBorrowerActions.sol	8027c146cec3cf24a30cd913228e12985e4755fd4d4ac8ead06cfc27b629eb87
open-term-loan-private/contracts/MapleLoan.sol	0f268dd43e01619eaae7886e04591e01035ea43734752719791bdb7ff4deca4f
fixed-term-loan-private/contracts/MapleLoan.sol	0582cb2ed70a9a40da356008058061bca62562f13fa34f7370b4498f026d4b9d
maple-core-v2-private/scripts/BorrowerActionsDeployment.s.sol	ecf68c712bac4a91eac657afaa0c8f741534b74f6e31acabc9357f3d3b56445e

Source Code	SHA256
maple-core-v2- private/scripts/BorrowerActionsProxyDepl oyment.s.sol	3dd6d249673253ff24a99a9a4c9ff7723 5195b97b7a64282ce8f6b05e3787654

Note: This document contains an audit solely of the Solidity contracts listed above. Specifically, the audit pertains only to the contracts themselves, and does not pertain to any other programs or scripts, including deployment scripts.

Issue Descriptions and Recommendations

Click on an issue to jump to it, or scroll down to see them all.

- ➦ Missing functionality for allowing BorrowerActions to call rejectNewTerms() for a fixed-term loan

Security Level Reference

We quantify issues in three parts:

1. The high/medium/low/spec-breaking **impact** of the issue:

- How bad things can get (for a vulnerability)
- The significance of an improvement (for a code quality issue)
- The amount of gas saved (for a gas optimization)

2. The high/medium/low **likelihood** of the issue:

- How likely is the issue to occur (for a vulnerability)

3. The overall critical/high/medium/low **severity** of the issue.

This third part – the severity level – is a summary of how much consideration the client should give to fixing the issue. We assign severity according to the table of guidelines below:

Severity	Description
(C-x) Critical	We recommend the client must fix the issue, no matter what, because not fixing would mean significant funds/assets WILL be lost.
(H-x) High	We recommend the client must address the issue, no matter what, because not fixing would be very bad, or some funds/assets will be lost, or the code's behavior is against the provided spec.
(M-x) Medium	We recommend the client to seriously consider fixing the issue, as the implications of not fixing the issue are severe enough to impact the project significantly, albeit not in an existential manner.
(L-x) Low	The risk is small, unlikely, or may not be relevant to the project in a meaningful way. Whether or not the project wants to develop a fix is up to the goals and needs of the project.
(Q-x) Code Quality	The issue identified does not pose any obvious risk, but fixing could improve overall code quality, on-chain composability, developer ergonomics, or even certain aspects of protocol design.
(I-x) Informational	Warnings and things to keep in mind when operating the protocol. No immediate action required.
(G-x) Gas Optimizations	The presented optimization suggestion would save an amount of gas significant enough, in our opinion, to be worth the development cost of implementing it.

Issue Details

✚✚

Missing functionality for allowing BorrowerActions to call rejectNewTerms() for a fixed-term loan

TOPIC	STATUS	IMPACT
Best practices	Fixed 	Informational *

In the `MapleLoan` contract of the fixed-term-loan-private repository, `BorrowerActions` actor has been added to act on behalf of the borrower. However, `rejectNewTerms()` has not been updated, and the necessary functionality to allow the `BorrowerActions` actor to execute this operation is missing. Since the `MapleBorrowerActions` contract does not yet support this functionality, it does not introduce any issues. However, for consistency, it should follow a similar logic from the open-term `MapleLoan` contract.

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The scope of this report and review is limited to a review of only the code presented by the Maple Finance team and only the source code Macro notes as being within the scope of Macro's review within this report. This report does not include an audit of the deployment scripts used to deploy the Solidity contracts in the repository corresponding to this audit. Specifically, for the avoidance of doubt, this report does not constitute investment advice, is not intended to be relied upon as investment advice, is not an endorsement of this project or team, and it is not a guarantee as to the absolute security of the project. In this report you may through hypertext or other computer links, gain access to websites operated by persons other than Macro. Such hyperlinks are provided for your reference and convenience only, and are the exclusive responsibility of such websites' owners. You agree that Macro is not responsible for the content or operation of such websites, and that Macro shall have no liability to your or any other person or entity for the use of third party websites. Macro assumes no responsibility for the use of third party software and shall have no liability whatsoever to any person or entity for the accuracy or completeness of any outcome generated by such software.